

EB-JEPA — Robustness, Hyperparameter Tuning & Ablation (V1)

EB-JEPA EEG track

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Five experiments distilled from a 5-stage optimization of our EB-JEPA on TUAB (v3.0.1, patient-disjoint, 276 eval recordings; frozen encoder $\rightarrow$ logistic-regression probe). Row 1 is the **tuned model** — the winner of the campaign (capacity $\rightarrow$ augmentation $\rightarrow$ schedule $\rightarrow$ loss coefficients $\rightarrow$ architecture/data); rows 2–5 each perturb it along one axis. *per-rec* = per-recording (16-window mean-pool, the clinical metric); *per-win* = per-window (the literature convention). Single-seed. Companion to `reference_table_JEPA_V1` (the full runs table).

#	Test	Configuration (vs. tuned model)	Params	per-rec BAcc / AUROC	per-win BAcc / AUROC
1	Tuned model	Conv · VICReg + spectral 0.1 + corruption · inv = 1 · no scale-jitter · 20 ep	1.35M	0.824 / 0.896	0.755 / 0.839
1b	Tuned — 3-seed	same recipe, seeds {0, 1, 2} (mean $\pm$ std)	1.35M	0.812 $\pm$ 0.021 / 0.898 $\pm$ 0.002	0.756 $\pm$ 0.001 / 0.836 $\pm$ 0.005
2	Robustness	pretrain on 25% of recordings (4 $\times$ less data)	1.35M	0.816 / 0.895	0.762 / 0.842
3	HP tuning	invariance weight inv = 25 (VICReg default) instead of 1	1.35M	0.804 / 0.884	0.730 / 0.813
4	Ablation	remove the corruption mask (outliers only)	1.35M	0.796 / 0.891	0.762 / 0.841
5	Ablation	Transformer encoder at equal capacity	1.29M	0.725 / 0.804	0.652 / 0.708

Legend / fixed setup. Every row uses the tuned model’s recipe except the stated change: a 0.4M-class 1-D conv scaled to **1.35M** (hidden=96, depth=4); two-view VICReg energy + a DDSP multi-scale-spectral auxiliary term (weight 0.1); exact EB-corruption views v_1, v_2 (Gaussian noise $\sigma = 0.1$, channel-drop $p = 0.2$, 20% timepoints masked + 20% $\pm 6\sigma$ outliers); invariance weight 1; 20 epochs; frozen-encoder logreg probe.

What each row shows.

- 1b Seed robustness (important caveat):** the per-recording BAcc carries $\approx \pm 0.02$ seed noise (only 276 recordings, threshold-sensitive) — seed 1 dipped to 0.788. Per-window BAcc (± 0.001) and AUROC (± 0.002) are stable. *Consequence: the per-rec Δ ’s in rows 2–4 (< 3 pp) sit within this noise band — only row 5’s -10 pp is statistically unambiguous. Rank configs on per-window / AUROC, not single-seed per-rec.*
- 2 Robustness to data:** cutting pretraining data 4 $\times$ costs only 0.8 pp (0.824 $\rightarrow$ 0.816) — the encoder is *not* data-starved; it saturates early.
- 3 Hyperparameter tuning:** VICReg’s vision-default invariance weight (25) is **2 pp worse** than our tuned inv=1. EEG SSL prefers *weak* view-invariance (full swept curve: 1 > 5 > 10 > 25).
- 4 Augmentation ablation:** the corruption **mask is the single biggest lever** (-2.8 pp when removed); for contrast, scale-jitter and the spectral term are *dead weight* (≈ 0).
- 5 Architecture ablation:** a from-scratch Transformer at *equal* ~ 1.3 M params is **10 pp worse** — at our scale/epoch budget the conv wins decisively; the SOTA gap is pretraining corpus + schedule, not architecture.